

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Limits

Week 5 Day 1 PM



OUTLINE

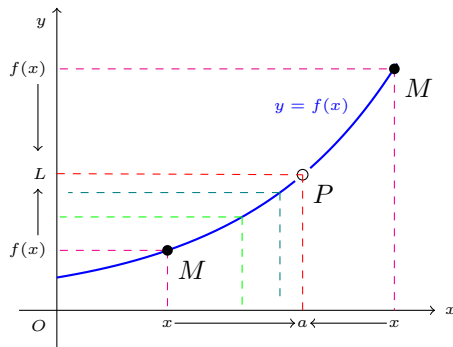
① Limits of Functions at a Finite Value

② One Sided Limits of Functions

③ Continuity

④ Summary

INTUITION OF LIMITS OF FUNCTIONS AT A FINITE VALUE



If $f(x) \rightarrow L$ when $x \rightarrow a$, we say that the limit of $f(x)$ is L when x approaches a .

$$x \rightarrow a \Rightarrow M \rightarrow P \Rightarrow f(x) \rightarrow L$$

Intuition: The function $f(x)$ approaches L as x approaches a .

WHY LIMITS?

Suppose $s(t)$ is the displacement of a particle at time t s and we want to find its instantaneous velocity v at $t = 2$ s. We can estimate this by computing the average velocity over the time interval $[2, 2.1]$:

$$v(2) \approx \frac{s(2.1) - s(2)}{2.1 - 2}.$$

But this is not instantaneous enough, we further shrink the interval to $[2, 2.01]$:

$$v(2) \approx \frac{s(2.01) - s(2)}{2.01 - 2}.$$

Still, this is not instantaneous enough, we further shrink the interval to $[2, 2.001]$, $[2, 2.0001]$, In other words, we want to know what is the average velocity over the time interval $[2, t]$, when t approaches 2:

$$v(2) = \lim_{t \rightarrow 2} \frac{s(t) - s(2)}{t - 2}.$$

This is the idea of derivatives, which we'll learn next class.

WHY LIMITS?

Expressions such as $f(x) = \frac{\sin x}{x}$ and $g(x) = \sin\left(\frac{1}{x}\right)$ are defined for all real x except $x = 0$. Similarly, $h(x) = \left(1 + \frac{1}{x}\right)^x$, $k(x) = x^x$ and $\ell(x) = x^{\frac{1}{x}}$ are defined for $x > 0$ but not for $x = 0$.

Are there 'natural' values for these functions at $x = 0$? (Natural in the sense that the graph of the function is not broken.) If there are, is there a way to find these 'natural' values?

Yes! For instance, we can try computing $f(0.1)$, $f(0.01)$, $f(0.001)$, ... and see whether $f(x)$ approaches a value when $x \rightarrow 0$. If it does, then

the 'natural' value of $f(0) = \lim_{x \rightarrow 0} f(x)$.

However, we need a rigorous definition for limits instead of just saying $\lim_{x \rightarrow 0} f(x)$ is the value which $f(0.1)$, $f(0.01)$, $f(0.001)$, ... is approaching to. There's no end to this process and we can only guess the answer.

ACTIVITY 1

By using the idea from the previous slide, use Excel to find the following limits numerically, if they exist. Note that you should consider x 's approaching a both from the left and the right.

$$(a) \lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}.$$

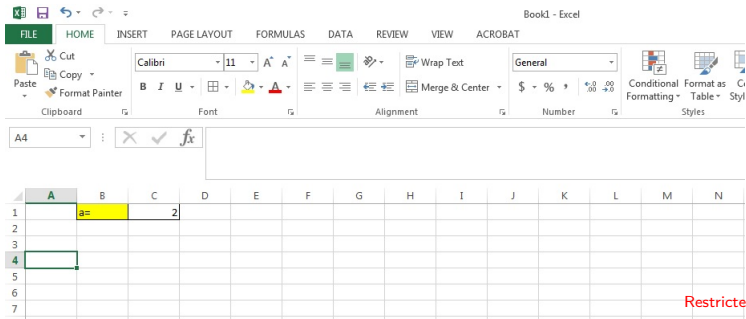
$$(b) \lim_{x \rightarrow 0} \frac{\sin x}{x}.$$

$$(c) \lim_{x \rightarrow \frac{\pi}{6}} \frac{\sin x - 0.5}{x - \frac{\pi}{6}}.$$

ACTIVITY 1: EXCEL GUIDES

To estimate $\lim_{x \rightarrow a} f(x)$, we look at the values of $f(x)$ at $x = a + 0.1$, $a + 0.01$, $a + 0.001$, ... (x approaching a from the right) and at $x = a - 0.1$, $a - 0.01$, $a - 0.001$, ... (x approaching a from the left). Excel can do these computations easily.

- (1) Set up the Excel sheet so that it's efficient, reader friendly and reusable. It's a good practice to create cells for the parameters such as a , so that we can easily make changes later if needed. Type $a=$ in cell B1 as a label and type 2 in cell C2, which is the a for problem 1.



ACTIVITY 1: EXCEL GUIDES

- (2) Create a column of $0.1, 0.01, 0.001, \dots$ (a geometric sequence) by typing 0.1 in a cell A4 and 0.01 in the cell below. Highlight the two cells and hover the cursor over the **Fill Handle** (lower right corner of the highlighted region). Press and hold the **right** mouse button, drag down the column, and release. A menu will appear. Click **Growth Trend** so that Excel will fill up the rest of the column to form a geometric sequence. (If you want an arithmetic sequence, just drag down with the **left** mouse button.)

The image shows three sequential screenshots of an Excel spreadsheet illustrating how to create a geometric sequence.

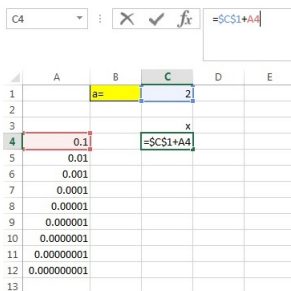
Left Screenshot: The spreadsheet has columns A, B, and C. Cell A4 contains the value 0.1, and cell A5 contains the value 0.01. Cells A4 and A5 are highlighted. The formula bar shows '0.1'.

Middle Screenshot: The right mouse button is pressed on the fill handle of the highlighted cells (A4:A5), opening a context menu. The menu options are: Copy Cells, Fill Series, Fill Formatting Only, Fill Without Formatting, Fill Days, Fill Weekdays, Fill Months, Fill Years, Linear Trend, **Growth Trend** (highlighted), Flash Fill, and Series....

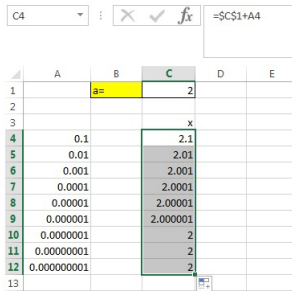
Right Screenshot: The 'Growth Trend' option has been selected. The cells A4 through A12 are now filled with the values of a geometric sequence: 0.1, 0.01, 0.001, 0.0001, 0.00001, 0.000001, 0.0000001, 0.00000001, and 0.000000001.

ACTIVITY 1: EXCEL GUIDES

- (3) To create a column of $2 + 0.1, 2 + 0.01, 2 + 0.001, \dots$, click on cell C4 and type $=\$C\$1+A4$, then hit enter. (the $=$ is to indicate that this is a formula instead of a string of letters, $\$C\$1+A4$ is to add the value in cells C1 and A4, the dollar signs in $\$C\1 are to fix the value in cell C1 as a constant, whenever we drag the formula to other cells.) Drag the lower right corner of cell C4 to the cells below to fill them up with the same formula applied to the corresponding input in column A. Increase the width of column C to see the actual values in the last few cells.



	A	B	C	D	E
1		a=	2		
2					
3			x		
4	0.1		=\\$C\\$1+A4		
5	0.01				
6	0.001				
7	0.0001				
8	0.00001				
9	0.000001				
10	0.0000001				
11	0.00000001				
12	0.000000001				
13					



	A	B	C	D	E
1		a=	2		
2					
3			x		
4	0.1		2.1		
5	0.01		2.01		
6	0.001		2.001		
7	0.0001		2.0001		
8	0.00001		2.00001		
9	0.000001		2.000001		
10	0.0000001		2		
11	0.00000001		2		
12	0.000000001		2		
13					

ACTIVITY 1: EXCEL GUIDES

- (4) Similarly, create a column of $2 - 0.1, 2 - 0.01, 2 - 0.001, \dots$ in column C.
- (5) Type $(x^3 - 8)/(x - 2)$ in cell D3 to label that column as values of $\frac{x^3 - 8}{x - 2}$. **Do not** put = as this is just a label, not a formula.
- (6) In cell D4, calculate the value of $\frac{x^3 - 8}{x - 2}$ at $x = 2.1$ by typing $=(C4^3 - 8)/(C4 - 2)$ and hit enter.

	A	B	C	D	E
1		a=	2		
2					
3				x $(x^3 - 8)/(x - 2)$	
4		0.1	2.1	$=(C4^3 - 8)/(C4 - 2)$	
5		0.01	2.01		
6		0.001	2.001		
7		0.0001	2.0001		
8		0.00001	2.00001		
9		0.000001	2.000001		
10		0.0000001	2.0000001		
11		0.00000001	2.00000001		
12		0.000000001	2.000000001		
13			1.9		
14			1.99		
15			1.999		
16			1.9999		

ACTIVITY 1: EXCEL GUIDES

- (7) Drag the lower right corner of cell D4 to the cells below to fill them up with the same formula applied to the corresponding input in column C. You can also do this by double clicking the lower right corner of cell D4. If you want to change the precision of the values, select the cells, right click to get a menu, choose **Format Cells**, select **Number** under the category and change the **Decimal places** to 15. Again, increase the width of the column to see the actual values. So, can you guess the value of $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$?

	A	B	C	D
1		a=	2	
2				
3			x (x^3-8)/(x-2)	
4	0.1	2.1		12.61
5	0.01	2.01		12.0601
6	0.001	2.001		12.006001
7	0.0001	2.0001		12.00060001
8	0.00001	2.00001		12.00006
9	0.000001	2.000001		12.000006
10	0.0000001	2.0000001		12.0000006
11	0.00000001	2.00000001		12
12	0.000000001	2.000000001		12
13		1.9		11.41
14		1.99		11.9401
15		1.999		11.994001
16		1.9999		11.99940001
17		1.99999		11.99994
18		1.999999		11.999994
19		1.9999999		11.9999994
20		1.99999999		12
21		1.999999999		12



	A	B	C	D
1		a=	2	
2				
3			x (x^3-8)/(x-2)	
4	0.1	2.1		12.61
5	0.01	2.01		12.0601
6	0.001	2.001		12.006001
7	0.0001	2.0001		12.00060001
8	0.00001	2.00001		12.00006
9	0.000001	2.000001		12.000006
10	0.0000001	2.0000001		12.0000006
11	0.00000001	2.00000001		12
12	0.000000001	2.000000001		12
13		1.9		11.41
14		1.99		11.9401
15		1.999		11.994001
16		1.9999		11.99940001
17		1.99999		11.99994
18		1.999999		11.999994
19		1.9999999		11.9999994
20		1.99999999		12
21		1.999999999		12

ACTIVITY 1: EXCEL GUIDES

(8) Do the same for problem 2 and 3. Use $\sin()$ for the sine function, $\text{pi}()$ for π .

	A	B	C	D	E	F	G	H	I	J
1		a=	2			0			0.523598776	
2										
3			x	$(x^3-8)/(x-2)$		x	$\sin(x)/x$		x	$(\sin(x)-0.5)/(x-\pi/6)$
4	0.1		2.1	12.610000000000000						
5	0.01		2.01	12.060100000000000						
6	0.001		2.001	12.006001000000900						
7	0.0001		2.0001	12.000600009997200						
8	0.00001		2.00001	12.000060000182600						
9	0.000001		2.000001	12.000006000533400						
10	0.0000001		2.0000001	12.000000603961300						
11	0.00000001		2.00000001	12.000000000000000						
12	0.000000001		2.000000001	12.000000000000000						
13			1.9	11.410000000000000						
14			1.99	11.940100000000000						
15			1.999	11.994001000000100						
16			1.9999	11.9994000100004500						
17			1.99999	11.99994000000083900						
18			1.999999	11.999993999466600						
19			1.9999999	11.999999396038700						
20			1.99999999	12.000000000000000						
21			1.999999999	12.000000000000000						
22										

Activity 1

Although it's easy to pick up Excel, it has its limitations. In SUTD, you'll learn more serious programming languages like Python.

ACTIVITY 1: SOLUTION

- (a) By substituting $x = 2.1, 2.01, 2.001, \dots$ and $x = 1.9, 1.99, 1.999, \dots$, the expression approaches 12, so the limit should be 12.
- (b) By substituting $x = 0.1, 0.01, 0.001, \dots$ and $x = -0.1, -0.01, -0.001, \dots$, the expression approaches 1, so the limit should be 1.
- (c) By substituting $x = \frac{\pi}{6} + 0.1, \frac{\pi}{6} + 0.01, \frac{\pi}{6} + 0.001, \dots$ and $x = \frac{\pi}{6} - 0.1, \frac{\pi}{6} - 0.01, \frac{\pi}{6} - 0.001, \dots$, the expression approaches $0.8660254 \dots \approx \frac{\sqrt{3}}{2}$, so the limit should be $\frac{\sqrt{3}}{2}$.

Note that this is not a rigorous way of finding limits. We are not 100% sure that the above limits are correct as there are a gazillion ways to approach a other than $a \pm (0.1)^n$ and we only look at a finite number of values. That's why we need to define limits of functions rigorously.

DEFINITION OF LIMIT OF FUNCTIONS AT A FINITE VALUE

DEFINITION[†]:

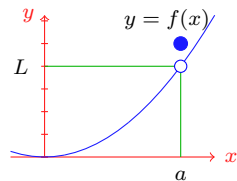
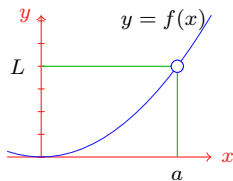
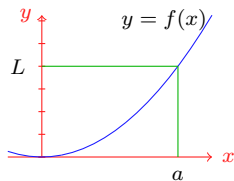
Intuitively, we say that $f(x)$ **converges to** $L \in \mathbb{R}$ **when x tends to/approaches** a , and denote this as $\lim_{x \rightarrow a} f(x) = L$, if the distance between $f(x)$ and L is arbitrarily small as long as x is close enough to a , but not equal to a (i.e. whatever happens at $x = a$ doesn't matter).

In this case, it can be shown that such a value L is unique and we say that the **limit** of $f(x)$ is L at $x = a$ /when x tends to/approaches a or sometimes we will just say that $\lim_{x \rightarrow a} f(x)$ **exists** without specifying the L .

[†] This is not the actual definition, the actual one is technical and abstract. It's called the ε - δ definition of limit.

⚠ Note that the limit of $f(x)$ at $x = a$ is **not** the value of f evaluated at $x = a$.

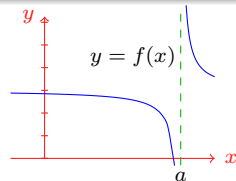
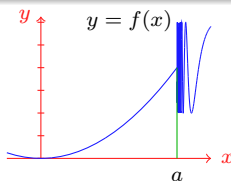
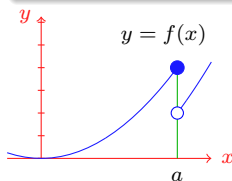
Try this [Geogebra applet](#).

EXAMPLES OF $\lim_{x \rightarrow a} f(x) = L$ 

DEFINITION OF DIVERGENCE

DEFINITION:

If $f(x)$ does not converge to **any** $L \in \mathbb{R}$ when x approaches a , we say that $\lim_{x \rightarrow a} f(x)$ **doesn't exist** or $f(x)$ **diverges** when x approaches/tends to a .



DEFINITION:

If $f(x)$ gets arbitrarily large (resp. negative large) as x tends to a , we say that $f(x)$ **diverges to ∞ (resp. $-\infty$)** and denote this as

$$\lim_{x \rightarrow a} f(x) = \infty.$$

$$(\text{resp. } \lim_{x \rightarrow a} f(x) = -\infty.)$$

RULES OF COMPUTING LIMITS

THEOREM

Suppose $\lim_{x \rightarrow a} f(x) = L_1$ and $\lim_{x \rightarrow a} g(x) = L_2$ (L_1, L_2 can be $\pm\infty$) and let $c \in \mathbb{R}$ be a constant. Then

- $\lim_{x \rightarrow a} c = c,$
- $\lim_{x \rightarrow a} x = a,$
- $\lim_{x \rightarrow a} cf(x) = c \left(\lim_{x \rightarrow a} f(x) \right) = cL_1,$
- $\lim_{x \rightarrow a} (f(x) \pm g(x)) = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x) = L_1 \pm L_2,$
- $\lim_{x \rightarrow a} (f(x) \cdot g(x)) = \lim_{x \rightarrow a} f(x) \times \lim_{x \rightarrow a} g(x) = L_1 \cdot L_2,$
- $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L_1}{L_2},$

with the same conventions and indeterminate forms as limits of sequences.

RULES OF COMPUTING LIMITS - EXAMPLE

E.g.:

$$\begin{aligned}\lim_{x \rightarrow -1} 7 \left(\frac{x^2 - 1}{x + 1} \right) &= 7 \lim_{x \rightarrow -1} \frac{(x + 1)(x - 1)}{x + 1} \\ &= 7 \lim_{x \rightarrow -1} (x - 1) = 7(-1 - 1) = -14.\end{aligned}$$

E.g.:

$$\begin{aligned}\lim_{x \rightarrow 2} \left(\frac{x + 1}{x - 1} - x^3 \right) &= \lim_{x \rightarrow 2} \frac{x + 1}{x - 1} - \lim_{x \rightarrow 2} x^3 \\ &= \frac{\lim_{x \rightarrow 2} (x + 1)}{\lim_{x \rightarrow 2} (x - 1)} - \left(\lim_{x \rightarrow 2} x \right) \left(\lim_{x \rightarrow 2} x \right) \left(\lim_{x \rightarrow 2} x \right) \\ &= \frac{2 + 1}{2 - 1} - 2 \cdot 2 \cdot 2 = 3 - 8 = -5.\end{aligned}$$

ACTIVITY 2

Find the following limits, if they exist.

(a) $\lim_{x \rightarrow 1} \frac{x - 4}{x^2 + 2}.$

(b) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}.$

ACTIVITY 2: SOLUTION

(a)

$$\begin{aligned}\lim_{x \rightarrow 1} \frac{x - 4}{x^2 + 2} &= \frac{\lim_{x \rightarrow 1} (x - 4)}{\lim_{x \rightarrow 1} (x^2 + 2)} \\&= \frac{\lim_{x \rightarrow 1} x - \lim_{x \rightarrow 1} 4}{\lim_{x \rightarrow 1} x \lim_{x \rightarrow 1} x + \lim_{x \rightarrow 1} 2} \\&= \frac{1 - 4}{1 \times 1 + 2} = -1.\end{aligned}$$

(b)

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2} &= \lim_{x \rightarrow 2} \frac{(x - 2)(x^2 + 2x + 4)}{x - 2} \\&= \lim_{x \rightarrow 2} (x^2 + 2x + 4) \quad (\text{because } x \neq 2) \\&= 2^2 + 2(2) + 4 = 12.\end{aligned}$$

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LEFT AND RIGHT LIMITS

It's possible to just consider x approaching a from one side only, in which case the limit is a one sided limit.

DEFINITION:

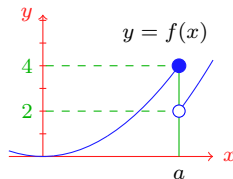
We say that $f(x)$ **converges to L when x approaches a from the left** and we write $\lim_{x \rightarrow a^-} f(x) = L$ if the distance between $f(x)$ and L is arbitrarily small as long as x is close enough to a and $x < a$. $\lim_{x \rightarrow a^-} f(x)$ is called the **left limit of $f(x)$ at $x = a$** .

DEFINITION:

We say that $f(x)$ **converges to L when x approaches a from the right** and we write $\lim_{x \rightarrow a^+} f(x) = L$ if the distance between $f(x)$ and L is arbitrarily small as long as x is close enough to a and $x > a$. $\lim_{x \rightarrow a^+} f(x)$ is called the **right limit of $f(x)$ at $x = a$** .

LEFT AND RIGHT LIMITS - EXAMPLE

E.g.:



$$\lim_{x \rightarrow a^-} f(x) = 4$$

and

$$\lim_{x \rightarrow a^+} f(x) = 2.$$

THEOREM ON LEFT AND RIGHT LIMITS

THEOREM

$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x).$$

In other words, the function $f(x)$ converges to a real number L if and only if $f(x)$ converges to L both with x approaches a from the left and when x approaches a from the right.

The theorem also provides us means to show that f diverges:

COROLLARY:

- If $f(x)$ does not converge to any $L \in \mathbb{R}$ as $x \rightarrow a^-$ or a^+ , then $f(x)$ diverges as $x \rightarrow a$.
- If $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$, then $f(x)$ diverges as $x \rightarrow a$.

ACTIVITY 3

Recall that the absolute value function is

$$|x| := \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Does $\frac{|x|}{x}$ converge as $x \rightarrow 0$? If so, what is $\lim_{x \rightarrow 0} \frac{|x|}{x}$? If not, why?

Notice that $\frac{|x|}{x}$ is not defined at $x = 0$. Does this matter in computing $\lim_{x \rightarrow 0} \frac{|x|}{x}$?

ACTIVITY 3: SOLUTION

For x approaching 0 from the left, $|x| = -x$ and we have

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = -1.$$

For x approaching 0 from the right, $|x| = x$ and we have

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = 1.$$

Since $\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1 \neq 1 = \lim_{x \rightarrow 0^+} \frac{|x|}{x}$, so $\frac{|x|}{x}$ diverges as $x \rightarrow 0$.

By the definition of limit, the value of $\lim_{x \rightarrow a} f(x)$ doesn't depend on the value of $f(a)$ (whatever happens at $x = a$ doesn't matter). So it doesn't matter whether $\frac{|x|}{x}$ is defined at $x = 0$ in computing

$$\lim_{x \rightarrow 0} \frac{|x|}{x}.$$

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DEFINITION OF CONTINUOUS FUNCTIONS

DEFINITION:

We say that f is **continuous at $x = a$** if a is in the domain of f and $f(x)$ converges to $f(a)$ when x approaches a , i.e.

$\lim_{x \rightarrow a} f(x) = f(a)$. Otherwise, f is **discontinuous at a** /has a **discontinuity at a** .

⚠ Continuity and limits are different concepts, but they are related. There are 3 conditions in the definition of continuity:

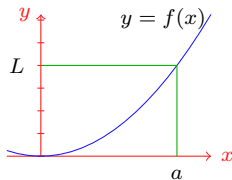
- $f(a)$ is defined (i.e. a is in the domain of f),
- $\lim_{x \rightarrow a} f(x)$ exists,
- $\lim_{x \rightarrow a} f(x) = f(a)$.

DEFINITION:

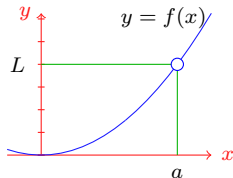
A function $f(x)$ is **continuous** if it is continuous at every point in its domain.

In layman terms, a continuous function is a nice function such that computing its limit at $x = a \in \text{domain}$ is to simply evaluate it at $x = a$.

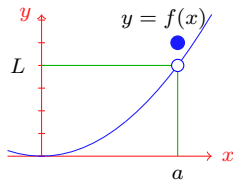
CONTINUOUS FUNCTIONS



$f(x)$ is continuous
at $x = a$, because
 $f(a) = \lim_{x \rightarrow a} f(x)$



$f(x)$ is not continuous
at $x = a$, because f is
not defined at $x = a$



$f(x)$ is not continuous
at $x = a$, because
 $f(a) \neq L = \lim_{x \rightarrow a} f(x)$

PROPERTIES OF CONTINUOUS FUNCTIONS

We'll state some important and useful properties of continuous functions here without proof.

PROPERTIES OF CONTINUOUS FUNCTIONS

- If $f(x)$ and $g(x)$ are continuous at a point a , then $f(x) + g(x)$, $f(x) - g(x)$ and $f(x)g(x)$ are also continuous at a . Furthermore $\frac{f(x)}{g(x)}$ is also continuous at a if $g(a) \neq 0$.
- Any polynomial $P(x)$ is continuous at all $x \in \mathbb{R}$.
- Any rational function $R(x) = \frac{P(x)}{Q(x)}$, where $P(x)$ and $Q(x)$ are polynomials, is continuous at all values of x for which $Q(x) \neq 0$.
- Exponential functions $f(x) = C^x$, logarithmic functions $g(x) = \log_C x$, $h(x) = x^C$, and all trigonometric functions $\sin, \cos, \tan, \csc, \sec, \cot$ are continuous, i.e. they are continuous at all points in their domain.

PROPERTIES OF CONTINUOUS FUNCTIONS

THEOREM

If $f(t)$ is continuous at $t = \lim_{x \rightarrow a} g(x)$, then

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right).$$

In particular, we have

- $\lim_{x \rightarrow a} x^n = a^n$ for any $n \in \mathbb{N}$ and $a \in \mathbb{R}$.
- $\lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}$ for any $n \in \mathbb{N}$ and $a > 0$.

LIMITS AND CONTINUOUS FUNCTIONS

What's the use of the previous two slides?

If you know that $f(x)$ is continuous using slide 30, then computing $\lim_{x \rightarrow a} f(x)$ **for a in the domain** of f is a piece of cake! You just need to substitute $x = a$ into $f(x)$ to compute the limit!

E.g.:

$$\begin{aligned}\lim_{x \rightarrow \pi} \left[\left(\frac{x^2 - 1}{x + 2} \right)^{\frac{1}{3}} + e^x \cos x \right] &= \left(\frac{\pi^2 - 1}{\pi + 2} \right)^{\frac{1}{3}} + e^{\pi} \cos \pi \\ &= \left(\frac{\pi^2 - 1}{\pi + 2} \right)^{\frac{1}{3}} - e^{\pi}.\end{aligned}$$

ACTIVITY 4

Find the following limits, if they exist.

$$(a) \lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h}$$

$$(b) \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{2}{x^2 + 2x} \right)$$

$$(c) \lim_{x \rightarrow \infty} \cos^2 x$$

$$(d) \lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$$

$$(e) \lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2}$$

Hint: for problems involving $\sqrt{\quad}$, consider the conjugate radical.

ACTIVITY 4: SOLUTION

(a)

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h} &= \lim_{h \rightarrow 0} \frac{9 + 6h + h^2 - 9}{h} \\ &= \lim_{h \rightarrow 0} \frac{6h + h^2}{h} \\ &= \lim_{h \rightarrow 0} (6 + h) = 6.\end{aligned}$$

(b)

$$\begin{aligned}\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{2}{x^2 + 2x} \right) &= \lim_{x \rightarrow 0} \frac{(x+2) - 2}{x^2 + 2x} \\ &= \lim_{x \rightarrow 0} \frac{x}{x^2 + 2x} \\ &= \lim_{x \rightarrow 0} \frac{1}{x+2} = \frac{1}{2}.\end{aligned}$$

ACTIVITY 4: SOLUTION

(c) By considering the sequence $(x_n = 2n\pi)_{n=0}^{\infty}$ where $\lim_{n \rightarrow \infty} x_n = \infty$, we have

$$\lim_{n \rightarrow \infty} \cos^2(x_n) = \lim_{n \rightarrow \infty} \cos^2(2n\pi) = \lim_{n \rightarrow \infty} 1^2 = 1.$$

On the other hand, by considering the sequence $(u_n = 2n\pi + \frac{\pi}{2})_{n=0}^{\infty}$ where $\lim_{n \rightarrow \infty} u_n = \infty$, we have

$$\lim_{n \rightarrow \infty} \cos^2(u_n) = \lim_{n \rightarrow \infty} \cos^2\left(2n\pi + \frac{\pi}{2}\right) = \lim_{n \rightarrow \infty} 0^2 = 0.$$

Since we have found two sequences $(x_n)_{n=0}^{\infty}$ and $(u_n)_{n=0}^{\infty}$ such that $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} u_n = \infty$, and

$$\lim_{n \rightarrow \infty} \cos^2(x_n) = 1 \neq 0 = \lim_{n \rightarrow \infty} \cos^2(u_n),$$

hence the limit doesn't exist.

ACTIVITY 4: SOLUTION

(d)

$$\begin{aligned}\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4} &= \lim_{x \rightarrow 4} \left(\frac{\sqrt{x} - 2}{x - 4} \cdot \frac{\sqrt{x} + 2}{\sqrt{x} + 2} \right) \\&= \lim_{x \rightarrow 4} \frac{(\sqrt{x})^2 - 2^2}{(x - 4)(\sqrt{x} + 2)} \\&= \lim_{x \rightarrow 4} \frac{x - 4}{(x - 4)(\sqrt{x} + 2)} \\&= \lim_{x \rightarrow 4} \frac{1}{\sqrt{x} + 2} = \frac{1}{\sqrt{4} + 2} = \frac{1}{4}.\end{aligned}$$

ACTIVITY 4: SOLUTION

(e)

$$\begin{aligned}\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2} &= \lim_{t \rightarrow 0} \left(\frac{\sqrt{t^2 + 9} - 3}{t^2} \cdot \frac{\sqrt{t^2 + 9} + 3}{\sqrt{t^2 + 9} + 3} \right) \\&= \lim_{t \rightarrow 0} \frac{(\sqrt{t^2 + 9})^2 - 3^2}{t^2(\sqrt{t^2 + 9} + 3)} \\&= \lim_{t \rightarrow 0} \frac{t^2 + 9 - 9}{t^2(\sqrt{t^2 + 9} + 3)} \\&= \lim_{t \rightarrow 0} \frac{1}{\sqrt{t^2 + 9} + 3} = \frac{1}{\sqrt{9} + 3} = \frac{1}{6}.\end{aligned}$$

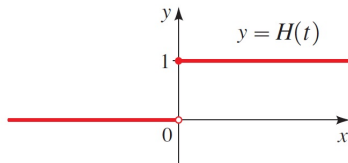
DISCONTINUOUS FUNCTIONS

You might think that all the functions used in engineering are continuous, but this is far from being true!

E.g.: In Laplace transform and circuit analysis, the Heaviside step function $H(x)$ plays an important role.

$$H(x) := \begin{cases} 0, & \text{for } x < 0, \\ 1, & \text{for } x \geq 0. \end{cases}$$

$H(x)$ has a discontinuity at $x = 0$.



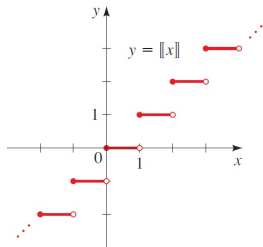
DISCONTINUOUS FUNCTIONS

E.g.: The greatest integer/floor function $\lfloor x \rfloor$ or $\llbracket x \rrbracket$ is used to round down numbers to integers:

$$\lfloor \cdot \rfloor : \mathbb{R} \longrightarrow \mathbb{Z}$$

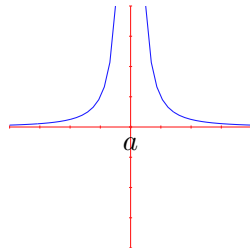
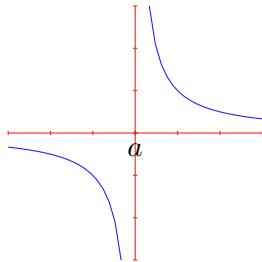
$$x \longmapsto \lfloor x \rfloor := \text{greatest integer less than or equal to } x.$$

It has discontinuities at all integers x .



DISCONTINUITY

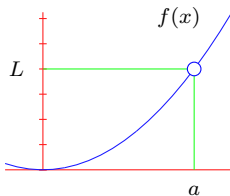
When $f(x)$ is discontinuous at $x = a$, the graph of $f(x)$ has a 'break' at $x = a$. We can categorize the discontinuity into the following:



$$\lim_{x \rightarrow a^-} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \rightarrow a^+} f(x) = \pm\infty$$

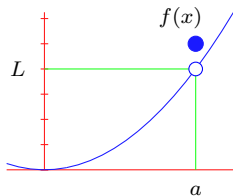
Infinite discontinuity

DISCONTINUITY



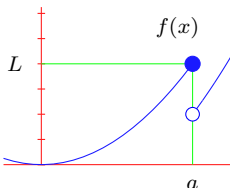
$\lim_{x \rightarrow a} f(x)$ exists but $f(a)$ is not defined

Removable discontinuity



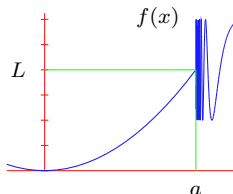
$f(a) \neq \lim_{x \rightarrow a} f(x)$

Removable discontinuity



$\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$

Jump discontinuity



$\lim_{x \rightarrow a} f(x)$ doesn't exist

Essential discontinuity

OUTLINE

① Limits of Functions at a Finite Value

② One Sided Limits of Functions

③ Continuity

④ Summary

SUMMARY

- Limits of Functions at a Finite Value
 - the intuition and motivation.
 - definition of limit of functions at a finite value, convergence and divergence.
 - rules of computing limits
- One Sided Limits of Functions
 - left and right limits
 - theorem on left and right limits
- Continuity
 - definition and properties of continuous functions.
 - limits and continuous functions.
 - discontinuous functions and different types of discontinuity.

RESOURCES

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 13.1, 13.2.